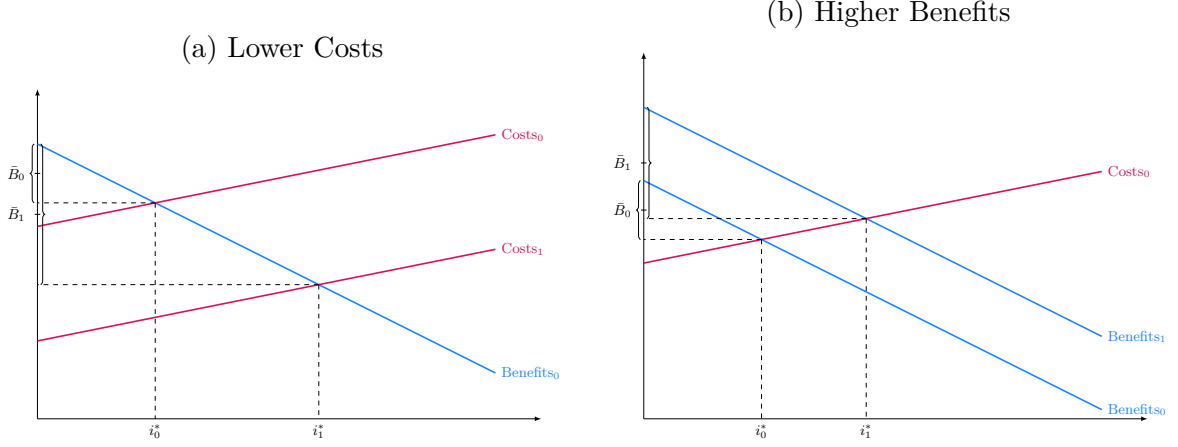


Figure A.1: The Impact of Employer Initiatives on Adoption Rates and Reported Benefits



Notes: This figure illustrates how employer initiatives affect adoption rates (i^*) and average reported benefits among users ($\bar{B}$). Panel (a) shows the impact of initiatives that reduce adoption costs (α_c), while Panel (b) depicts the impact of initiatives that enhance benefits (α_b). Lower adoption costs (α_c) unambiguously increase adoption rates and reduce average reported benefits due to a negative selection effect. In contrast, initiatives that raise benefits (α_b) also increase adoption, but their impact on average reported benefits is theoretically ambiguous and, at most, equal to their individual-level effect.

In summary, the average reported benefits among users provide a lower bound on the causal effect of employer initiatives on *individual* benefits, due to the presence of a negative *selection* effect.

In this light, the fact that Table 1 shows higher reported benefits among users in encouraged settings is striking. It suggests two things: (i) employer encouragement primarily raises perceived benefits (α_b), as reductions in adoption costs (α_c) would only generate the negative selection effect; and (ii) the employer-induced increases in benefits outweigh the adverse selection effect—implying that employer encouragement is powerful relative to the degree of individual heterogeneity ($\gamma < 1$).

By contrast, the finding that enterprise chatbots and training—when implemented in isolation—are associated with lower reported benefits may reflect either that these initiatives reduce benefits for all users ($\alpha_{b1} < 0$), or that negative selection effects outweigh any positive individual-level gains.

A.1.3 Discussion

The preceding section examines the implications of workers self-selecting into chatbot adoption based on their individual benefits. Beyond this margin of selection, at least two additional forms of selection may be at play.

First, firms that choose to encourage chatbots may be the ones with the greatest benefits from the tools—even absent any encouragement initiatives—introducing potential reverse causality between employer encouragement and reported benefits. In the framework above, this corresponds to E being positively related to α_{b0} . Second, employers may target their encouragement toward employees who are expected to benefit the most individually. This would resemble a “rotation” of the cost and benefit schedules in Figure A.1, rather than simple vertical shifts.

Section E addresses both of these additional selection margins empirically. First, Section E.4 examines whether employers select into encouragement based on their underlying benefits from chatbot use. We show that the effects of encouragement remain robust when controlling for a range of firm- and worker-level characteristics, including firm age, size, and productivity, as well as workers’ detailed task mixes and other attributes. These robustness checks show that the association between employer encouragement and reported benefits is not driven by observable confounders.

Second, Section E.5 implements a “coworker encouragement” design, using the average encouragement reported by a worker’s colleagues to measure workplace-wide encouragement rates. Because coworker encouragement rates capture only *workplace*-level variation, this approach helps address potential bias arising from employers targeting encouragement toward *individual* employees with the highest idiosyncratic benefits. As shown in Table E.14 and Figure E.21, the findings remain largely robust when using coworker encouragement as the source of variation.

A.2 Labor Market Outcomes

This section provides a theoretical analysis of how AI chatbots affect labor markets, followed by a discussion of how these effects can be identified empirically.

Modeling the labor market impacts of AI chatbots raises several key questions: How do chatbots enter the production process? What explains why some firms and workers adopt them while others do not? Are labor markets competitive, or do firms exercise monopsony power? How do firms compete in product markets? Before committing to a formal framework, Section A.2.1 considers the theoretical implications of each of these issues.

Section A.2.2 then introduces a fully-fleshed model of chatbot adoption and labor market outcomes. We embed the Roy adoption model from Section A.1 into a simple equilibrium framework in which heterogeneous firms adopt chatbot initiatives and hire workers to maximize profits, while workers sort across firms and adopt chatbots to maximize utility. Section A.2.4 uses the model to derive predictions for how AI chatbots affect labor market outcomes. Finally, Section A.2.5 applies the model to interpret our difference-in-differences estimates from Section 3.

For clarity of exposition, we focus on encouraged use as the employer initiative, reflecting its central role in the empirical analyses in Section 3.

A.2.1 Modeling Choices

Chatbot technology. A first key question is how chatbots enter the production process: Are they “co-pilots,” contributing to output only when prompted by humans, or closer to “co-workers,” operating autonomously without human involvement?²¹ More broadly, do AI chatbots complement or substitute for labor?

Acemoglu and Restrepo (2018*b*, 2019) provide a theoretical framework for how new technologies affect labor productivity through three channels: a negative *task displacement effect*, where chatbots automate tasks previously performed by humans (e.g., automated

²¹Ide and Talamas (2025) embed AI in a model of the knowledge economy, showing that whether AI functions as a co-pilot or co-worker has crucial implications for its impact on labor demand and inequality.

copyediting); a positive *task reinstatement effect*, where new tasks for humans emerge (e.g., oversight or integration of AI chatbots); and a positive *productivity effect*, where productivity improvements from AI chatbots enable firms to scale up production, including labor demand—depending on the elasticity of product demand. As they show, the adoption of new technologies (e.g., encouraging chatbot use) increases labor demand only when the combined productivity and reinstatement effects outweigh the displacement effect.

Modeling chatbots is further complicated by the fact that adoption is not solely a firm-level decision; it also depends on workers’ choices to use the tools. A key consideration is whether chatbot use augments or displaces workers’ skills. Several experiments suggest that chatbots benefit workers with less prior expertise, helping to close skill gaps within professions (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023). Following the worker-level adoption model in Section A.1, workers will use chatbots if the productivity benefits exceed the adoption costs. In particular, if the amenity value of chatbot use is sufficiently high, workers may adopt even when doing so reduces their productivity.

In sum, the implications of chatbot adoption for labor demand and productivity are theoretically ambiguous at both the firm and worker levels. At the firm level, the outcome depends on whether the productivity and reinstatement effects together outweigh the displacement effect. At the worker level, the productivity impact depends on the amenity value of chatbot use.

Firm and worker heterogeneity. Another key question is why some firms and workers adopt chatbots while others do not. Following the frameworks of Mas (2025); Rosen (1986), chatbots can be viewed as a “productive amenity”: firms will encourage chatbot use if the profit gains exceed their costs, while workers will adopt if the productivity benefits outweigh adoption costs (as in Section A.1). Applying the framework of Mas (2025), chatbots may thus induce a resorting of workers who prefer to use them into firms that encourage adoption, provided that encouragement enhances the utility of chatbot use (as found in Section 1.4). In competitive labor markets, Rosen (1986) shows that the wage